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Duc Nguyen

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Professional Summary

AI Engineer with expertise in software design, frontend/backend development, and cloud/Kubernetes management. Have designed and built mission critical AI research tools. Proficient across the AI lifecycle, from data acquisition and model building to deployment. Avid open-source contributor.

Professional Background

Independent Research **Oct 2024 - Now**

Objective: engage in frontier research on intelligence.

- Replicate research on LLM and Reinforcement Learning. Implement popular variants of LLM architecture: Qwen, DeepSeek, Llama. Implement LoRA fine-tuning and popular post-train RL algorithms: PPO, GRPO, DPO.
- Replicate research on mechanistic interpretability: Sparse Autoencoder, Crosscoder, Activation patching, Attribution Patching, Logit Lens, Transcoder, feature steering, and apply these techniques on non-transformer architectures: CNN, RNN, state space model, diffusion model.
- Refresh on math: calculus, linear algebra, probability & information theory. Obtained corresponding course certificates in these fields.
    - Linear Algebra from Elementary to Advanced Specialization
    - Information Theory
    - Advanced Calculus
- Improve on high performance scientific computing: numpy, numba, cuda, triton. Obtained course certificate for GPU Programming Specialization.

Founding Software Engineer **Amethyst Invest** **Oct 2024 - Now**

Objective: establish engineering foundation for quantitative trading R&D and operation.

- Build a research library for quantitative trading. Enable researchers to build and test new trading algorithms, and centralize algorithms for reuse. Allow researchers to scale up experiments, ensure trust and total reproducibility of research output.
- Optimize trading algorithm. Carefully vectorize algorithms with numpy and numba, to achieve minimum 20x speed up from research code.
- Grow team, guide researchers on algorithm optimization, data management, and engineering best practices.

AI Architecture Lead - RAG **Cinnamon AI Lab** **April 2023 – Oct 2024**

Objective: lead the design and development of a RAG-based Q&A product.

- Designed the architecture for KnowledgeHub, a RAG-based Q&A system on private data. Implemented core backend components: the database, the vector store, the search engine... Also created a smaller open-source version, available at https://github.com/Cinnamon/kotaemon.
- Built a scalable and cost-effective AI inference system on Kubernetes to serve private LLMs and other AI models for KnowledgeHub.
- Surveyed and implemented indexing-retrieval and agent pipelines from literature. Built a semi-automatic Q&A evaluation framework.
- Coordinated the improvement of indexing-retrieval and agent pipelines with team members.
- Fine-tuned private LLMs to enhance answering performance on clients' domain-specific data.

AI Architecture Lead - Data Management **Cinnamon AI Lab** **June 2020 – April 2023**

Objective: eliminate data-related problems in R&D.

- Designed and implemented Athena, the data management system, that encompasses the full data lifecycle: from data labeling, data validation, to data access and version control. Athena serves as a bedrock for all research and production projects at Cinnamon. It allows AI data to be centralized and indexed, so AI researchers can query and get the exact data for R&D. It allows data annotators to keep track of labeling progress and validate data quality.
- Designed and implemented the auto-labeling system in conjunction with Athena. It allows AI researchers to integrate their AI models to help data annotators automate the labeling job. This system expedites labeling progress by 8x. At the same time, it accelerates research iterations by providing AI researchers crucial insights on the actual usability of their AI models.
- Created the AI data workflow and applied the workflow to all AI researchers, engineers and data annotators.
- Built data schema for intelligence document processing. The data schema allows different research and production projects to agree on the same data interface. Without it, data will not be reusable.

AI OCR Team Lead **Cinnamon AI Lab** **Dec 2018 – June 2020**

- Led and coached team members directly to improve OCR performance and usability. Active research topics: generative adversarial network, semi-supervised learning, model pruning, text detection, language modeling, deep learning models reliability, robustness and interpretability.
- Researched and led the effort to combine document image transcription and document content understanding into one single process that reduces error accumulations and improves product processing time by 2.5 times.
- Established long-term OCR research roadmap and coordinated with team execution.
- Led the construction of machine learning pipelines for quick solution iteration and reproducible research.
- OCR Patent Pending: *Long Text Line Recognition using Convolutional Associative Memory Network*

AI Engineer **Cinnamon AI Lab** **Jan 2018 – Dec 2018**

- Developed handwriting text line OCR models that achieve state-of-the-art performance on Chinese and Japanese handwriting recognition.
- Devised image synthesizing approach that generates data with great sensible variations for OCR model training.
- Followed closely clients' requests for product improvements.

Full-Stack Developer **ACC Consultancy** **Sept 2016 – Dec 2017**

- Designed and implemented both backend and front-end for https://cpp.community that serves more than 1000 daily unique visitors.
- Benchmarked and enhanced website security.
- Streamlined the process to collect, clean and update data for the website.
- Worked closely with supervisor and designer to enhance the user experience.

EDUCATION

New London, NH **Colby – Sawyer College** **Aug 2011 – May 2015**

- B.S in Philosophy
- GPA: 3.9 / 4.00 – Magna Cum Laude

OTHERS

Personal Awards:

- 2021 KDD Document Intelligence Workshop Best Paper Award for the paper *HYCEDIS: HYbrid Confidence Engine for Deep Document Intelligence System*
- 2018 1st place AWS Hackathon Vietnam
- 2019 Obstacle Tower Challenge Finalists

Technical Skills

- Programming Languages: Python, JavaScript, C++, C, CUDA, Triton, Go
- Machine Learning Frameworks: PyTorch, TensorFlow
- Cloud Platforms: AWS, Azure, GCP
- DevOps Tools: Docker, Kubernetes and its stacks (Knative, Keda, cluster auto-scaler…), Github Actions, CircleCI
- Data Technologies: PostgreSQL, MySQL, MongoDB, sqlite, ElasticSearch, Redis, Memcached, Neo4j
- Frontend Technologies: React, Svelte
- Backend Technologies: Django, FastAPI, Litestar, Gunicorn, uvicorn

Personal Side Projects:

- **Dawnet** (2018 - ongoing): visualize and inspect inner-workings of deep learning models.
- **Reproduce Paper Quick & Dirty** (2018 - ongoing): follow and reimplement notable deep learning articles.
- **god** (2021 - ongoing): a git-inspired, distributed data version control system.
- **theflow** (2023 - ongoing): a composable pipeline execution framework that can run each component cross-platform.